TOM LACLAVÈRE

Doctor in Astrophysics · Software Engineer – Numerical Simulation & HPC

Doctor in astrophysics and software engineer, specialized in numerical simulation and performance optimization. Lead developer of simulation and data reconstruction pipelines for the QUBIC experiment, deployed on HPC clusters. Seeking a position in numerical simulation or HPC.

PROFIL

- tomlaclavere@gmail.com
- 0643382043
- Permis B/A2
- TomLaclavere
- Website
- LinkedIn

SKILLS

- Languages: C++ (advanced), Python (advanced), Fortran (intermediate)
- HPC: OpenMP, TBB, MPI (mpi4py), SLURM
  ↳ familiar with: SYCL, Thrust, Kokkos, CUDA
- Performance: Profiling (perf, Malt, Maqao), Optimisation, Vectorisation
- Tools: Git, CMake, Docker, Linux, CI/CD, unit testing.
- Data Analysis: Scipy, Pandas, sklearn, PyTorch, JAX

SOFT SKILLS

- Rigor and work organisation
- Autonomy and adaptability
- Written and oral communication
- Ability to explain complex technical concepts
- Technical ownership and sense of responsibility

LANGUAGES

- French (native)
- English (professional)
- Japanese (beginner)

HOBBIES

- Sports: Boxing, badminton
- Programming, video games, motorcycling, hiking

EXPERIENCE

2023-2026 — CNRS | Université Paris-Cité | APC Laboratory

PhD - QUBIC Experiment

- Lead developer of the simulation, reconstruction and data analysis pipeline (qubicsoft, > 30k LoC, > 500 commits).
- Large-scale linear system solving: numerical methods & machine learning.
- Parallelisation (mpi4py, OpenMP) and SLURM deployment (up to 20 nodes, 100-200 GB/run).
- Development of calibration and noise analysis tools for data validation.

2025- — GitHub Project

HPC / Simulation Project - 3DPhysicsEngine (C++23)

- 3D rigid body simulation engine: custom math library, two-phase collision pipeline, three integrators (Euler, Verlet, RK4)
- Benchmarks: 150+ configurations (solver × time step), convergence, performance and energy drift quantified
- Current phase: CPU optimization (SIMD, OpenMP); roadmap: CUDA/SYCL, MPI

EDUCATION

2023-2026 — **PhD in Fundamental Physics - Astrophysique**

- QUBIC Data Analysis: Realistic astrophysical components reconstruction and atmospheric mitigation using spectral imaging (APC)
- Goal: Constrain inflation parameters by detecting primordial CMB B-modes, separating astrophysical foregrounds and atmospheric contamination via spectral imaging.

2025 — **Gray Scott Summer School - HPC (CC-FR/LAPP)**

- TBB, OpenMP, OpenMPI, SYCL, CUDA, Thrust
- C++, Fortran, Rust, Python, Julia

2023 — **Master Nuclei, Particles, Astroparticles & Cosmology (NPAC)** - Mention bien (Université Paris-Cité)

2021 — **Bachelor's in Fundamental Physic** - Mention Très bien (Université Paris-Cité)

2018-2020 — **CPGE, Physics-Chemistry** - Admitted to Centrale (Cité Scolaire Bertran-de-Born)

TEACHING

2024 - 2026 — **Adjunct Teaching Assistant**

CentraleSupélec
- Cosmology - 1ère année - Tutorials (35h) - 2026

Université Paris-Cité
- Physique Numérique - Master 1 - Tutorials (36h) - 2024 & 2025

Ecole d'Ingénieur Denis Diderot
- Champs Electromagnétique - 1ère année - Tutorials (24h) - 2024
- Bruits - 2ème année - Lab sessions (24h) - 2024